

The General Theory of Lumped Electrical Devices

A Unified State-Space, Graph-Theoretic, Energy-Based, and Impedance-Theoretic Framework

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Abstract

This paper develops a general mathematical framework for electrical and electronic devices constructed from arbitrary combinations of resistive, inductive, and capacitive elements. Using graph theory, state-space analysis, network theory, impedance methods, energy methods, and port-Hamiltonian systems, we derive a universal representation of finite lumped electrical systems. Connections are drawn to physics, chemistry, biology, economics, finance, psychology, psychiatry, political science, international relations, warfare economics, artificial intelligence, computer science, and systems theory. The resulting framework demonstrates that every finite lumped RLC device is a finite-dimensional dynamical system admitting a rational transfer function representation.

The paper ends with “The End”

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1 Introduction

Electrical devices constitute one of the most successful examples of complex systems assembled from simple components. A resistor, capacitor, and inductor each obey comparatively simple constitutive laws. Nevertheless, their interconnection generates oscillators, filters, communication systems, computers, control systems, and modern digital civilization.

The objective of this paper is to derive the most general mathematical representation of a finite electrical or electronic device constructed from arbitrary series and parallel combinations of resistance, inductance, and capacitance.

2 Primitive Elements

Consider a network containing

$$N_R$$

resistors,

$$N_L$$

inductors,
and

$$N_C$$

capacitors.

2.1 Resistors

Ohm's Law gives

$$v_{R_k} = R_k i_{R_k}.$$

2.2 Inductors

Faraday's law implies

$$v_{L_k} = L_k \frac{di_{L_k}}{dt}.$$

2.3 Capacitors

The capacitor law is

$$i_{C_k} = C_k \frac{dv_{C_k}}{dt}.$$

2.4 Generalized Components

The most general constitutive relations may be nonlinear:

$$R_k = R_k(x, t),$$

$$L_k = L_k(x, t),$$

$$C_k = C_k(x, t).$$

3 Graph-Theoretic Representation

3.1 Network Graph

Define a graph

$$G = (V, E),$$

where

- vertices correspond to nodes,
- edges correspond to circuit elements.

Let

$$A$$

denote the incidence matrix.

3.2 Kirchhoff Current Law

Kirchhoff Current Law becomes

$$Ai = 0.$$

3.3 Kirchhoff Voltage Law

Kirchhoff Voltage Law becomes

$$A^T v = v_s.$$

Theorem 1. *Every finite lumped electrical network admits a graph-theoretic representation.*

Proof. Every electrical component connects exactly two nodes (or can be decomposed into such connections). Therefore every component can be represented as an edge and every junction as a vertex. Consequently the network defines a graph. Kirchhoff's laws become incidence matrix constraints. $\square$

4 State-Space Representation

Define the state vector

$$x = \begin{bmatrix} i_L & v_C \end{bmatrix}.$$

The state dimension is

$$n = N_L + N_C.$$

The general system becomes

$$\dot{x} = f(x, u, t).$$

For linear systems,

$$M\dot{x} + Kx = Bu.$$

where

$$M = \begin{bmatrix} L & 0 & 0 & C \end{bmatrix}.$$

Thus

$$\dot{x} = -M^{-1}Kx + M^{-1}Bu.$$

5 Universal Impedance Representation

5.1 Laplace-Domain Elements

Resistor:

$$Z_R = R.$$

Inductor:

$$Z_L = sL.$$

Capacitor:

$$Z_C = \frac{1}{sC}.$$

5.2 Admittances

$$Y_R = \frac{1}{R},$$

$$Y_L = \frac{1}{sL},$$

$$Y_C = sC.$$

Therefore

$$Y_b(s) = \text{diag}(Y_1(s), Y_2(s), \dots, Y_m(s)).$$

The nodal admittance matrix is

$$Y(s) = A, Y_b(s), A^T.$$

6 Universal Rational Form

Theorem 2. *Every finite lumped linear RLC network possesses a rational impedance function.*

Proof. The constitutive equations are linear differential equations with constant coefficients. Applying the Laplace transform converts differentiation into multiplication by s . Solving the resulting algebraic system yields a ratio of finite-degree polynomials. $\square$

Thus

$$Z(s) = \frac{P(s)}{Q(s)}.$$

More explicitly,

$$Z(s) = \frac{a_0 + a_1 s + \dots + a_n s^n}{b_0 + b_1 s + \dots + b_m s^m}.$$

7 Energy Theory

7.1 Magnetic Energy

$$E_L = \frac{1}{2} \sum_{k=1}^{N_L} L_k i_k^2.$$

7.2 Electric Energy

$$E_C = \frac{1}{2} \sum_{k=1}^{N_C} C_k v_k^2.$$

7.3 Total Energy

$$E = E_L + E_C.$$

7.4 Dissipation

$$P_D = \sum_{k=1}^{N_R} R_k i_k^2.$$

Proposition 1. *The energy balance equation is*

$$\frac{dE}{dt} = P_{\text{in}} - P_D.$$

8 Port-Hamiltonian Formulation

Define the Hamiltonian

$$H(x) = E(x).$$

Then

$$\dot{x} = (J - R)\nabla H(x) + Gu.$$

where

$$J = -J^T$$

and

$$R = R^T \geq 0.$$

9 Connections to Physics

The framework directly incorporates

1. Maxwell's Equations,
2. Electromagnetic Energy Conservation,
3. Wave Propagation,
4. Resonance Phenomena,
5. Quantum Circuit Models.

Electrical circuits may be viewed as finite-dimensional reductions of distributed electromagnetic field theories.

10 Connections to Chemistry

Electrochemical systems obey analogous storage and flow laws:

Electrical Quantity	Chemical Analogue
Voltage	Chemical Potential
Current	Reaction Flux
Capacitance	Storage Capacity
Resistance	Reaction Impedance

Table 1: Chemical analogues.

Batteries, fuel cells, and electrolytic systems can therefore be interpreted within the generalized network framework.

11 Connections to Biology

Neurons exhibit electrical dynamics similar to RLC systems.

Examples include:

- Hodgkin-Huxley dynamics,
- Membrane capacitance,
- Ion-channel resistance,
- Synaptic filtering.

Biological signaling networks may therefore be interpreted as generalized electrical circuits.

12 Economics

Economic systems exhibit accumulation, dissipation, and flow.

Circuit Quantity	Economic Analogue
Charge	Wealth
Current	Cash Flow
Resistance	Transaction Cost
Capacitance	Savings Capacity
Inductance	Institutional Inertia

Table 2: Economic analogues.

The dynamic equation

$$M\dot{x} + Kx = Bu$$

appears naturally in macroeconomic adjustment processes.

13 Finance

Financial markets display oscillatory behavior analogous to damped RLC systems.

Examples include:

- Yield-curve oscillations,
- Liquidity storage,
- Market friction,
- Momentum effects.

The transfer function framework provides a natural frequency-domain interpretation of financial dynamics.

14 Psychology and Psychiatry

Psychological systems can be viewed as networks possessing:

- Memory storage,
- Feedback loops,
- Dissipative mechanisms,
- Oscillatory behavior.

Capacitive effects correspond to memory retention while resistive effects correspond to forgetting and damping.

Circuit Concept	Political Analogue
Node	State
Edge	Relationship
Current	Resource Flow
Resistance	Friction
Capacitance	Strategic Reserve
Inductance	Institutional Momentum

Table 3: Political analogues.

15 Political Science and International Relations

States may be interpreted as nodes connected by economic, military, informational, and diplomatic channels.

16 Warfare Economics

Military logistics networks satisfy generalized flow equations analogous to electrical networks.

Examples include

- Fuel supply systems,
- Ammunition distribution,
- Personnel deployment,
- Strategic reserves.

The existence of storage, transport, and dissipation naturally produces RLC-like structures.

17 Computer Science

Modern computers are physical realizations of large-scale electrical networks.

Applications include

- Logic circuits,
- Signal processing,
- Network routing,
- Distributed systems.

18 Artificial Intelligence

Neural networks may be interpreted as generalized dynamical systems.

The state-space framework

$$\dot{x} = f(x, u, t)$$

appears naturally in

- Recurrent Neural Networks,
- Neural ODEs,
- State Space Models,
- Continuous-Time Learning Systems.

19 Statistical Interpretation

Suppose uncertainty affects component values.

Then

$$R_k \sim F_R,$$

$$L_k \sim F_L,$$

$$C_k \sim F_C.$$

The impedance becomes a random variable:

$$Z(s, \omega).$$

Reliability analysis follows from the resulting stochastic system.

20 Algorithmic Construction

Pseudo-Code

Input:

Circuit Graph G

Construct Incidence Matrix A

For each branch:

Determine R, L, C parameters

Build $Y_b(s)$

Compute:

$$Y(s) = A Y_b(s) A^T$$

Solve:

$$I(s) = Y(s)V(s)$$

Return:

Transfer Function

State-Space Model

Energy Function

21 Illustrative Figure

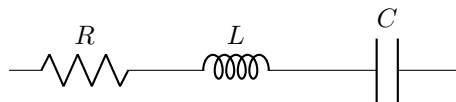


Figure 1: Simple series RLC network as a special case of the universal framework.

22 Comparative Table

Field	Storage	Flow	Dissipation
Electrical	Capacitance	Current	Resistance
Economics	Savings	Spending	Friction
Biology	Memory	Signaling	Decay
Military	Reserves	Logistics	Attrition
AI	State	Information	Regularization

Table 4: Cross-disciplinary analogies.

23 Main Result

Theorem 3 (Universal Lumped Device Representation). *Every finite electrical device composed of resistive, inductive, and capacitive components admits a representation*

$$M(x, t)\dot{x} + K(x, t)x = B(x, t)u,$$

and, in the linear case,

$$Z(s) = \frac{P(s)}{Q(s)}.$$

Proof. The result follows from graph-theoretic representation, Kirchhoff constraints, constitutive equations, and Laplace-domain transformation. Finite-dimensionality guarantees polynomial numerator and denominator representations. $\square$

24 Conclusion

The most general finite electrical or electronic device can be represented simultaneously as a graph, a state-space system, an energy-conserving dynamical system, a port-Hamiltonian system, and a rational impedance operator. This unifies circuit theory with broader theories of dynamical systems across the physical, biological, social, economic, military, and computational sciences.

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Glossary

A	Incidence matrix.
B	Input matrix.
C	Capacitance.
E	Stored energy.
G	Circuit graph.
$H(x)$	Hamiltonian.
J	Interconnection matrix.
K	System matrix.
L	Inductance.
M	Mass-like storage matrix.
R	Resistance.
$Y(s)$	Admittance matrix.
$Z(s)$	Impedance function.
x	State vector.

The End